

Effectivity

All LightSpeed QX/i Systems manufactured before Jan 1, 2001 with Gantry Model Number 2180482 and loaded with Rev 1.1, 1.2, and preliminary 1.3 software releases (run "showprods" to determine software version).

showprods <ENTER>

<Helios_CLASSC_Product 11/09/2000 Helios OC Class C Product 10.5.1_2.8.2I_H1.3M4

The above line shows the Product already at 1.3M4.

Some of the systems near this manufacturing cut in date may have the new revision of Software loaded. IF the system has LightSpeed QX/i Applications Software Version **10.5.x_x.x.xx_H1.3M4**, the FMI is not required. Otherwise, the FMI is applicable.

Also, It is possible to combine FMIs 25263 and 25269 to go directly from Version 1.1 to Version 1.3 and save doing a software load of Version 1.2. However, in this case the hardware modification must be done from the Version 1.2 upgrade (FMI 25263).

Purpose

This FMI is the method of distribution being used to distribute the upgrades and bug fixes provided by this new release of software. The features introduced with this new software include (but are not limited to) the following:

- Thin Twin Axial Mode (2 x .63 mm)
- Multi-Destination DICOM Print Capability
- Latest release of Inste Interactive (IIP 2.1)
- Advanced Artifact Reduction (AAR)
- Web Browser Based Common Service Desktop - Phase II
 - Includes a new HomePage
 - New Log Viewer
- Separated Class A, C, and M software
- Beam-Tracking Improvements

Prerequisite FMI's

All Open LightSpeed FMI's must be installed on the system prior to this FMI. This includes FMI# 25261 (LightSpeed 1.1 Z-Axis Tracking) and FMI# 25263 (LightSpeed Version 1.2 Upgrade). However, if going directly from Version 1.1 to Version 1.3, only the hardware modifications of the Version 1.2 upgrade (FMI 25263) need to be done. **If going directly from Ver 1.1 to Ver 1.3,**

you must have FMI 25263 (Ver 1.2 Upgrade kit) on-hand for both the instructions and the hardware.

Related FMI's

FMI 25263, Version 1.2 Upgrade

Furnished Materials

FMI Kit Part Number 2286980 includes the items listed below

Qty	Description	Part #
1	FMI Document Rev 1	2282388-100
1	LightSpeed QX/i Rev 1.3 System Software Collector	2260334-5
	consisting of 1.3 Software CD-ROM Collector 2258825-7 and Software Rating Plate	
	LightSpeed QX/i Rev 1.3 System Service Manual General Rev 0	2282383-100
1	LightSpeed QX/i Rev 1.3 System Installation Manual General Rev 1	2282384-100
1	LightSpeed QX/i Technical Reference Manual, Rev 5	2211214-100
1	LightSpeed QX/i Rev 1.3 Load From Cold Manual Rev 0	2282382-100
1	LightSpeed QX/i Sherlock Tutorial CDROM, Rev 4	2226999
1	LightSpeed 1.3 Service Introduction Planner Rev 2	2282387-100
1	LightSpeed QX/i Applications Tips and Work Arounds, Rev 5	2211223-100

Tools Required

MOD for Save State

Poly and QA Phantoms that are on each site

ILinq Package (Distributed on Tab 283)

ILinq Software Version 10 (GE Part Number 2243504-7)

ILinq Installation Manual (Direction 2243323-100, Rev 00 or higher)

Class C Software Version 10(GE Part Number 2284160-2).

LightSpeed QX/i Rev 1.3 Advanced System Service Manual (2282386-100)

LightSpeed QX/i Rev 1.3 Advanced System Installation Manual (2282385-100)

People Power

Labor Estimates: 6 hours

Travel Estimates: 1 hour

Procedure

Introduction

This FMI includes the following activities:

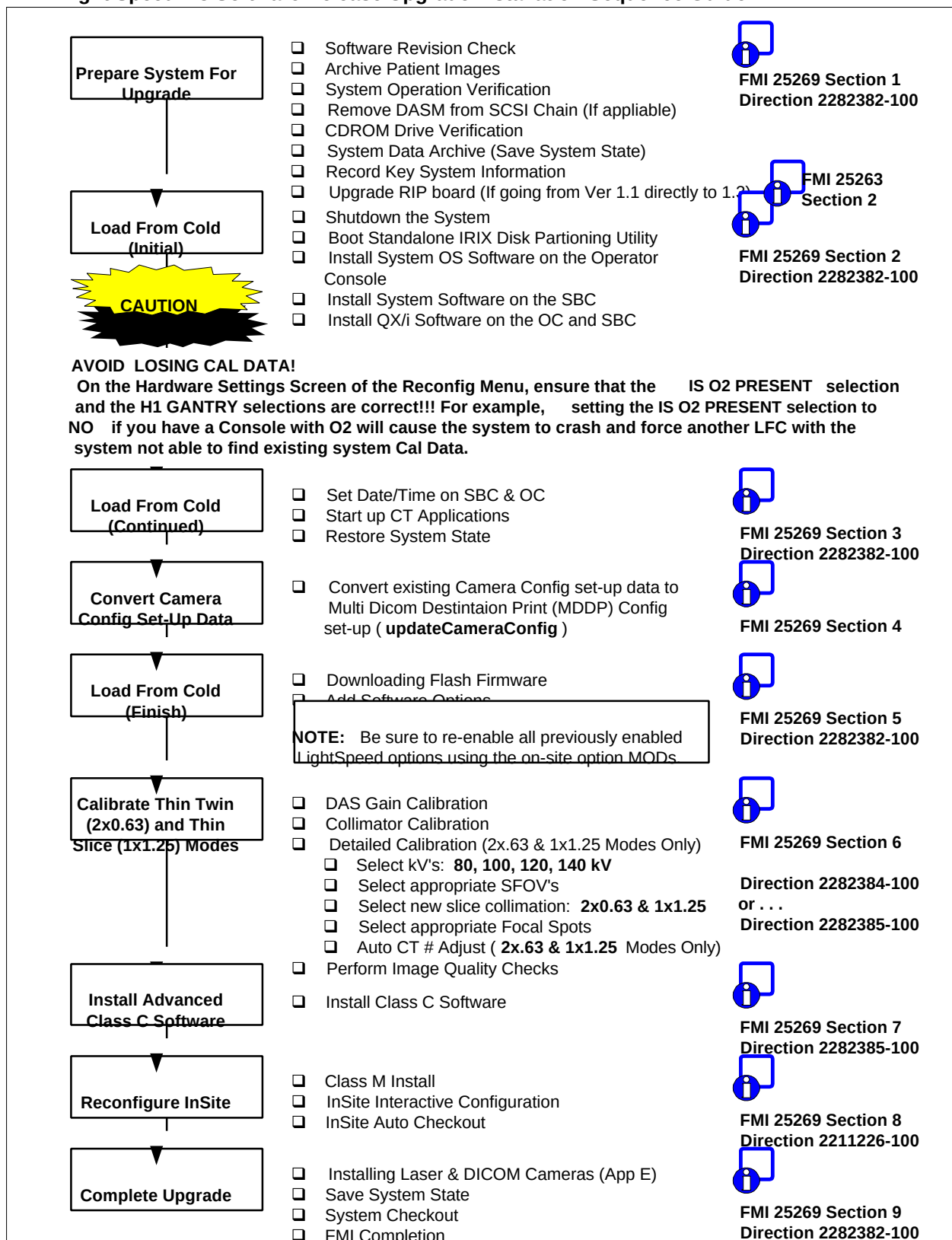
- System Data Archive
- Upgrade RIP board per FMI 1.2, if needed (If upgrading system directly from Ver 1.1 to Ver 1.3)
- Load From Cold (LFC)
- System Calibrations
- DICOM Reconfiguration
- IQ Checks
- Save System State
- System Checkout
- FMI Completion

Install Sequence Guide

Refer to Figure 1 for a LightSpeed 1.3 Upgrade Installation Sequence Guide.

Figure 1

Light Speed 1.3 Software Release Upgrade Installation Sequence Guide



Section 1 Prepare System For Upgrade

Software Revision Check

The software revision can be check using the showprods command.

1.) Open a UNIX shell.

2.) Type:

```
showprods | grep -i "QXi Apps" <ENTER>
```

If you see <6.8.2_2.7.3n2> **!STOP! System is at 1.1 - FMI 25263 1.2, SECTION 2 MUST BE DONE FIRST**

Go to FMI 25263 direction and complete section 2 of that FMI until you finish the RIP Board Upgrade.

Return to this FMI at this step and proceed after RIP Board Upgrade.

If you see <7.2.1_2.8.2F_HE1.2M3> or <8.12.2_2.8.2G_H1.2M4>, proceed with next step.

If you see <10.4.1_X.X.XX_H1.3M4> Proceed with this FMI - you have a preliminary version of 1.3 Software.

If you see <10.5.1_X.X.XX_H1.3M4> STOP - THIS FMI HAS ALREADY BEEN DONE.

Image Data Archive

The following procedures will destroy all image data on the system. Verify that the customer has reconstructed and archived all images before proceeding.

System Operation Verification

Setup and perform at least one axial scan to verify that the system operated properly before beginning the upgrade.

Remove DASM From SCSI Chain

During the load of applications software, the SCSI bus is being heavily utilized. During this process there have been sites that have experienced load problems. To minimize the occurrence of these problems, it is recommended to remove the DASM from the SCSI bus chain before continuing with this FMI. Once the LFC has been performed the DASM can be reinstalled.

To remove the DASM from the SCSI bus chain:

- 1.) At the SCSI Expansion box disconnect the cable coming from the DASM.
- 2.) At the DASM, remove the SCSI cable that comes from the Octane and connect it to the SCSI Expansion box connection opened in step one.

CDROM Drive SCSI BUS Verification

Since the CDROM Drive is seldom used except for loading system software, it is advisable at this time to perform a quick test on the ability of the CDROM drive to read data from a CD before you

begin the LFC procedure. This test is accomplished by inserting and mounting one of the OS CDROM disks, and performing a checksum (read check) on the files in one of the directories on the CD. If any errors occur while the system is reading this CDROM, you must stop to investigate and correct the CD problem (for example the SCSI Bus, CDROM Drive, etc. problems).

1.) Open a UNIX shell.

2.) Type:

```
tail -f /usr/adm/SYSLOG <ENTER>
```

3.) Open a second UNIX shell.

4.) Insert the LightSpeed QX/i **IRIX Operating System CDROM Disk #1** (p/n 2240216-2) into the CDROM Drive and type:

```
mountCD /mnt <ENTER> (Note: ~ 1 minute for the prompt to return)
```

```
cd /mnt/dist <ENTER>
```

```
sum * > /tmp/CDROMtest <ENTER> (~5 minutes to display prompt)
```

5.) While the system is reading the CDROM and calculating checksums, you need to monitor the SYSLOG in the first UNIX shell window for any error messages that may occur that are related to SCSI or CDROM problems (timeouts, read errors, SCSI Bus resets, etc.).

6.) If checks are successful (no errors produced), type the following command to verify the checksum:

```
sum /tmp/CDROMtest <ENTER>
```

7.) Verify that the resultant display for the command in step 6 is:

```
34499 8 /tmp/CDROMtest
```

8.) If you encounter any CDROM or SCSI related errors during this check, stop and troubleshoot/repair the problem before attempting to perform the FMI.

Troubleshooting suggestions included scoping the VME power supply outputs, check/replace the CDROM and/or SCSI cables.

9.) Terminate the 'tail' command by typing **^C** in the first UNIX shell window where the 'tail' command is running.

10.) Unmount the CDROM by typing the following commands in the second UNIX shell window (the same shell that you typed `cd /mnt/dist <ENTER>`):

```
cd / <ENTER>
```

```
unmountCD <ENTER>
```

System Data Archive (Save System State) and Record Key System Information

Refer to the LFC manual (2282382-100), Section 2.4 and 2.5 to save system cal files, protocols, characterization, etc. onto an MOD and make a record of the key system information.

Section 2 Load From Cold Procedure (Initial)

After system operation has been verified and data archiving has been completed, you are ready to upgrade the system software.

Using the LightSpeed QX/i Load From Cold Manual (2282382-100) and the software included in the FMI kit, perform a software load. **Refer to Section 2.6 through 2.11 of the LFC manual,** follow the procedure to load the software and restore the System State from MOD.

CAUTION: AVOID LOSING CAL DATA!!

On the Hardware Settings Screen of the **RECONFIG** menu, ensure that the **IS O2 PRESENT** selection is set appropriately (depending on whether your system is configured with a O2 computer or not).

This setting is only available once (at LFC time). Setting this selection incorrectly will cause the system to crash. This will force you to perform another Load From Cold with the system unable to find any existing calibration data.

Also, make sure you select H1 gantry. If you select H2, you will also crash and need to reload.

Section 3 Load From Cold Procedure (Continued)

Continue through the LFC process to Set date & time, start up CT Applications, and Restore System State. Refer to the LFC Manual (2282382-100), sections 2.12 to 2.14.

If any problems occur with the Flash Download, applications will have trouble performing Phantom Cals and scanning and the error Recon Subsystem Reset Failed will occur. If this happens, manually flash the RIP board using Addendum on page 10.

Section 4 Convert Camera Config to MDDP

The following instruction must be run to support Multi-Destination DICOM Printing. This is a one-time need to convert the camera configuration files. Actual camera set-up will be done in Section 9.

- Open a Unix shell (Service Desktop ==> Utilities ==> Shell).
- Type: updateCameraConfig
- After the unix prompt returns, close the shell window.

Section 5 Load From Cold Procedure (Finish)

CAUTION: FLASH FIRMWARE UPDATES REQUIRED (CCB/ DCB DOWNLOAD)

Flash download is required and will take 10-15 minutes. During this process there will be error messages indicating lack of communication. Please be patient while this process takes place. The "stop" key will go inactive when download is finished.

After the Flash Firmware Download is complete, add the software options back in using the original MODs found on-site for your customer.

Refer to the LightSpeed QX/i Load From Cold Manual (2282382-100), Sections 2.15 and 2.16.

Section 6 Calibrate 2 x .63 Thin Twin

System Calibration

1.) !STOP! Perform this step ONLY if you started with 1.1 software.

Go to next step if you started with 1.2 software

If you are upgrading from 1.1 software, you need to calibrate both 1 X 1.25 Single Slice & 2 X .63 Thin Twin. The recommendation from the engineering team is to step through the full set of calibrations and assure that each calibration impacted by 1 X 1.25 and by 2 X .63 is performed during the Detailed Calibration. Perform in order: **DAS Gain Cal., Collimator Cal., Detailed Cal.** (Air and Phantom Cals for 1 X 1.25 Single Slice & 2 X .63 Thin Twin) and **Auto CT # Adjust**. The longest this calibration should take will be 4.0 hours. For instructions see LightSpeed Installation Manual (2278881-100 or 2278883-100). Go to Section 7 now.

2.) Perform this step if you started with 1.2 software.

If you are upgrading from 1.2 software, you need to calibrate 2 X .63 Thin Twin. The recommendation from the engineering team is to step through the full set of calibrations and assure that each calibration impacted by 2 X .63 is performed during the Detailed Calibration. Perform in order: **DAS Gain Cal., Collimator Cal., Detailed Cal.** (Air and Phantom Cals for 2 X .63 Thin Twin) and **Auto CT # Adjust**. The longest this calibration should take will be 4.0 hours. For instructions see LightSpeed Installation Manual (2278881-100 or 2278883-100). Go to Section 7 now.

Section 7 Install Class - C Software

For this and all future versions, we'll be able to install or uninstall the proprietary software (Class - C) separately. This will provide us with more control of proprietary information.

Refer to the Advanced Installation Manual (2282385-100), Chapter 5, Section 3.4.1

Note: On the system, class - c install scripts are located in /usr/g/scripts

Section 8 Reconfigure InSite

Following the instructions in the InSite Manual (Direction 2211226-100), to load the InSite software, set up the InSite Interactive configuration, and perform auto checkout.

Section 9 Reconnect the DASM

Section 10 Complete Upgrade

Install Laser and DICOM Cameras

Multi-Destination DICOM Print is one of the new features of this release. Now that the system has this software, you need to ensure that the prior-Camera settings are correctly mapped into the new DICOM Configuration GUI. And, it is now possible to add additional Hospital DICOM Camera destinations to the system configuration.

Refer to the LFC Manual (2282382-100), Appendix - E.

Save System State

Refer to the LFC Manual, and perform a Save System State. This will save the new calibrations as well as the customer protocols.

System Checkout

Refer to the LFC Manual, and perform the system performance checks as described in the manual. Ensure all networking and filming processes are still functional.

Once system checkout has been performed, you can install the front cover of the console.

FMI Completion

Complete the Product Locator information cards and return.

Fill out and return the FMI Completion Certification Report paperwork (or electronically close your FMI dispatch) in accordance with your local procedure.

Addendum

Manual Flash RIP Firmware

During the LFC, the system performs a firmware flash to the RIP board. If the flash does not work properly, the system will fail phantom calibrations. If this happens, you may need to identify the cause of the fault and perform a manual pflash. To perform the firmware flash follow the steps below.

1. Shutdown the applications software:

Service Desktop

Tools/Utilities

Applications Shutdown

Wait 30 to 60 seconds for applications to shutdown before going to step 2

2. Open a Unix shell

3. Log into the SBC using the command:

rsh sbc <Enter>

4. Flash the firmware. Using the command:

/usr/g/ice/bin/pflash <Enter>

If flash works properly, you will see loading type text in approximately 12 lines.

5. Configure the Scan Data Disk using the command:

/usr/g/scripts/reconfigScanDisk <Enter>